COMICCRAFT – AI COMIC STORY CREATOR USING GEMINI MODELS

7. Project Documentation Phase

7.1 Introduction

The Project Documentation Phase records the complete information about the ComicCraft application.

Documentation helps developers, team members, faculty members, and users understand the purpose, structure, installation, working process, and features of the project.

7.2 Project Overview

ComicCraft is a Generative AI-based web application that uses Google Gemini models to create comic stories from user-provided inputs.

The application combines AI-based story generation, image generation, comic panel creation, and PDF export.

7.3 System Workflow

The general workflow of ComicCraft is:

User Input → Story Outline → Detailed Story → Image Prompts → Images → Comic Panels → Preview → PDF

The user first enters the required story information. The Gemini models process the information and generate the required story content.

7.4 Technology Documentation

The main technologies used in the project are:

- Python
- FastAPI
- HTML
- CSS
- JavaScript
- Google Gemini
- Hugging Face
- Jinja2
- Pillow

- FPDF2
- Pytest

7.5 Project Structure Documentation

The project contains different folders for different responsibilities.

The main folders include:

- `app` – application and backend files
- `services` – AI and processing services
- `templates` – HTML pages
- `static` – CSS, JavaScript, and generated content
- `tests` – testing files
- `docs` – project documentation

7.6 API Documentation

The important API routes are documented with their purpose, inputs, and expected outputs.

FastAPI also provides interactive API documentation through Swagger UI and ReDoc.

7.7 Configuration Documentation

The project uses environment variables for API keys and configuration values.

Important configuration values include:

- `GEMINI_API_KEY`
- `GEMINI_OUTLINE_MODEL`
- `GEMINI_STORY_MODEL`
- `HF_TOKEN`
- `HF_IMAGE_MODEL`

Sensitive API keys and tokens should not be exposed in a public repository.

7.8 User Documentation

A user can start the application, open the web interface, enter the required story information, generate the comic, preview the result, and export the completed comic as a PDF.

7.9 Conclusion

The Documentation Phase provides a complete reference for the ComicCraft project. It explains the project purpose, workflow, technologies, structure, configuration, API functionality, and usage process.